Year 11 Higher

REVISION PACK

Progress Check 1

ANSWERS

Units in this pack

- Number 8
- Algebra 7

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Q	Working	Answer
prime factors in index form		
1.1	$120 = 2 \times 60$ $60 = 2 \times 30$ $30 = 2 \times 15$ $15 = 3 \times 5$	$2^3 \times 3 \times 5$
1.2	$504 = 2 \times 252$ $252 = 2 \times 126$ $126 = 2 \times 63$ $63 = 3 \times 21$ $21 = 3 \times 7$	$2^3 \times 3^2 \times 7$
1.3	$600 = 2 \times 300$ $300 = 2 \times 150$ $150 = 2 \times 75$ $75 = 3 \times 25$ $25 = 5 \times 5$	$2^3 \times 3 \times 5^2$
1.4	$280 = 2 \times 140$ $140 = 2 \times 70$ $70 = 2 \times 35$ $35 = 5 \times 7$	$2^3 \times 5 \times 7$
1.5	$450 = 2 \times 225$ $225 = 3 \times 75$ $75 = 3 \times 25$ $25 = 5 \times 5$	$2 \times 3^2 \times 5^2$
1.6	$792 = 2 \times 396$ $396 = 2 \times 198$ $198 = 2 \times 99$ $99 = 3 \times 33$ $33 = 3 \times 11$	$2^3 \times 3^2 \times 11$
HCF and LCM		

1.7	**(a)** $90 = 2 \times 3^2 \times 5$ $150 = 2 \times 3 \times 5^2$ $\text{HCF} = 2 \times 3 \times 5 = 30$ **(b)** $\text{LCM} = 2 \times 3^2 \times 5^2 = 450$	**(a) 30** **(b) 450**
1.8	**(a)** $48 = 2^4 \times 3$ $180 = 2^2 \times 3^2 \times 5$ $\text{HCF} = 2^2 \times 3 = 12$ **(b)** $\text{LCM} = 2^4 \times 3^2 \times 5 = 720$	**(a) 12** **(b) 720**
1.9	**(a)** $72 = 2^3 \times 3^2$ $108 = 2^2 \times 3^3$ $\text{HCF} = 2^2 \times 3^2 = 36$ **(b)** $\text{LCM} = 2^3 \times 3^3 = 216$	**(a) 36** **(b) 216**
1.10	**(a)** $56 = 2^3 \times 7$ $84 = 2^2 \times 3 \times 7$ $\text{HCF} = 2^2 \times 7 = 28$ **(b)** $\text{LCM} = 2^3 \times 3 \times 7 = 168$	**(a) 28** **(b) 168**
1.11	**(a)** $126 = 2 \times 3^2 \times 7$ $168 = 2^3 \times 3 \times 7$ $\text{HCF} = 2 \times 3 \times 7 = 42$ **(b)** $\text{LCM} = 2^3 \times 3^2 \times 7 = 504$	**(a) 42** **(b) 504**
1.12	**(a)** $60 = 2^2 \times 3 \times 5$ $96 = 2^5 \times 3$ $\text{HCF} = 2^2 \times 3 = 12$ **(b)** $\text{LCM} = 2^5 \times 3 \times 5 = 480$	**(a) 12** **(b) 480**
next time together		
1.13	$\text{LCM of } 16 \text{ and } 24 = 48$ $7:00 + 48 \text{ minutes}$	**7:48 am**
1.14	$\text{LCM of } 10 \text{ and } 25 = 50$ $8:30 + 50 \text{ minutes}$	**9:20 am**

1.15	LCM of 14 and $21 = 42$ $6:00 + 42$ minutes	**6:42 am**
1.16	LCM of 20 and $30 = 60$ $9:15 + 60$ minutes	**10:15 am**
1.17	LCM of 18 and $24 = 72$ $7:30 + 72$ minutes	**8:42 am**
1.18	LCM of 12 and $15 = 60$ $10:00 + 60$ minutes	**11:00 am**

Number 8 · Standard form (Lesson 1.1)

Q	Working	Answer
calculating in standard form		
2.1	**(a)** $4 \times 2.5 = 1E + 1$ $10^{3+(-6)} = 10^{-3}$ 1×10^{-2} **(b)** $50\,000 + 3200 = 53\,200$	**(a)** 1×10^{-2} **(b)** 5.32×10^{4}
2.2	**(a)** $1.6 \times 5 = 8$ $10^{5+(-2)} = 10^{3}$ **(b)** $700\,000 + 65\,000 = 765\,000$	**(a)** 8×10^{3} **(b)** 7.65×10^{5}
2.3	**(a)** $2.4 \times 3 = 7.2$ $10^{-3+8} = 10^{5}$ **(b)** $2\,000\,000 + 850\,000 = 2\,850\,000$	**(a)** 7.2×10^{5} **(b)** 2.85×10^{6}
2.4	**(a)** $6 \times 1.5 = 9$ $10^{4+(-9)} = 10^{-5}$ **(b)** $4100 + 900 = 5000$	**(a)** 9×10^{-5} **(b)** 5×10^{3}
2.5	**(a)** $3.5 \times 4 = 14$ $10^{2+(-5)} = 10^{-3}$ 1.4×10^{-2} **(b)** $80\,000 + 2400 = 82\,400$	**(a)** 1.4×10^{-2} **(b)** 8.24×10^{4}

2.6	**(a)** $2.5 \times 8 = 2E + 1$ $10^{-4+7} = 10^3$ 2×10^4 **(b)** $120\,000 + 60\,000 = 180\,000$	**(a)** 2×10^4 **(b)** 1.8×10^5
writing numbers in standard form		
2.7	**(a)** 7.2 7.2×10^{-4} **(b)** $3.6 \div 10\,000 = 0.00036$	**(a)** 7.2×10^{-4} **(b) 0.00036**
2.8	**(a)** 4.15 4.15×10^{-5} **(b)** $5.2 \div 1000 = 0.0052$	**(a)** 4.15×10^{-5} **(b) 0.0052**
2.9	**(a)** 9.08 9.08×10^{-3} **(b)** $1.75 \div 100\,000 = 0.0000175$	**(a)** 9.08×10^{-3} **(b) 0.0000175**
2.10	**(a)** 6.3 6.3×10^{-5} **(b)** $8.04 \div 100 = 0.0804$	**(a)** 6.3×10^{-5} **(b) 0.0804**
2.11	**(a)** 1.23 1.23×10^{-2} **(b)** $4.9 \div 1\,000\,000 = 0.0000049$	**(a)** 1.23×10^{-2} **(b) 0.0000049**
2.12	**(a)** 2.08 2.08×10^{-7} **(b)** $6.3 \div 1000 = 0.0063$	**(a)** 2.08×10^{-7} **(b) 0.0063**

Number 8 · Ratio (Lesson 1.4)

Q	Working	Answer
sharing in a ratio		

3.1	$2+3+5=10$ $300 \div 10=30$ $2 \times 30=60$ $3 \times 30=90$ $5 \times 30=150$	**60 KD, 90 KD, 150 KD**
3.2	$1+2+4=7$ $420 \div 7=60$ $1 \times 60=60$ $2 \times 60=120$ $4 \times 60=240$	**60 KD, 120 KD, 240 KD**
3.3	$2+3+4=9$ $540 \div 9=60$ $2 \times 60=120$ $3 \times 60=180$ $4 \times 60=240$	**120 KD, 180 KD, 240 KD**
3.4	$3+4+5=12$ $600 \div 12=50$ $3 \times 50=150$ $4 \times 50=200$ $5 \times 50=250$	**150 KD, 200 KD, 250 KD**
3.5	$1+4+6=11$ $330 \div 11=30$ $1 \times 30=30$ $4 \times 30=120$ $6 \times 30=180$	**30 KD, 120 KD, 180 KD**
3.6	$1+3+5=9$ $720 \div 9=80$ $1 \times 80=80$ $3 \times 80=240$ $5 \times 80=400$	**80 KD, 240 KD, 400 KD**
ratio and difference		
3.7	$5-2=3 \text{ parts}$ $18 \div 3=6$ $7 \times 6=42$	**42**

3.8	$7 - 3 = 4$ parts $20 \div 4 = 5$ $10 \times 5 = 50$	**50**
3.9	$7 - 4 = 3$ parts $15 \div 3 = 5$ $11 \times 5 = 55$	**55**
3.10	$8 - 5 = 3$ parts $12 \div 3 = 4$ $13 \times 4 = 52$	**52**
3.11	$4 - 1 = 3$ parts $27 \div 3 = 9$ $5 \times 9 = 45$	**45**
3.12	$4 - 3 = 1$ parts $9 \div 1 = 9$ $7 \times 9 = 63$	**63**

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Q	Working	Answer
substitution		
4.1	$s = 2 \times 3 + \frac{1}{2} \times (-4) \times 3^2$ $= 6 - 18$ $= -12$	$s = -12$
4.2	$s = 6 \times 2 + \frac{1}{2} \times (-5) \times 2^2$ $= 12 - 10$ $= 2$	$s = 2$
4.3	$s = 4 \times 5 + \frac{1}{2} \times (-1.5) \times 5^2$ $= 20 - 18.75$ $= 1.25$	$s = 1.25$
4.4	$s = 1 \times 4 + \frac{1}{2} \times (3) \times 4^2$ $= 4 + 24$ $= 28$	$s = 28$

4.5	$s = 10 \times 2 + \frac{1}{2} \times (-2.5) \times 2^2$ $= 20 - 5$ $= 15$	$s = 15$
4.6	$s = 3 \times 6 + \frac{1}{2} \times (-2) \times 6^2$ $= 18 - 36$ $= -18$	$s = -18$
rearranging formulae		
4.7	**(a)** $y - 7 = 4x$ $x = \frac{y-7}{4}$ **(b)** $r^2 = \frac{A}{3}$ $r = \sqrt{\frac{A}{3}}$ **(c)** $5x - cx = 2 - b$ $x(5 - c) = 2 - b$ $x = \frac{2-b}{5-c}$	**(a)** $x = \frac{y-7}{4}$ **(b)** $r = \sqrt{\frac{A}{3}}$ **(c)** $x = \frac{2-b}{5-c}$
4.8	**(a)** $y + 9 = 2x$ $x = \frac{y+9}{2}$ **(b)** $r^2 = \frac{V}{\pi h}$ $r = \sqrt{\frac{V}{\pi h}}$ **(c)** $ax - 4x = b + 3$ $x(a - 4) = b + 3$ $x = \frac{b+3}{a-4}$	**(a)** $x = \frac{y+9}{2}$ **(b)** $r = \sqrt{\frac{V}{\pi h}}$ **(c)** $x = \frac{b+3}{a-4}$

4.9	**(a)** $y-1=6x$ $x=\dfrac{y-1}{6}$ **(b)** $v^2=\dfrac{2K}{m}$ $v=\sqrt{\dfrac{2K}{m}}$ **(c)** $7x-qx=3-p$ $x(7-q)=3-p$ $x=\dfrac{3-p}{7-q}$	**(a)** $x=\dfrac{y-1}{6}$ **(b)** $v=\sqrt{\dfrac{2K}{m}}$ **(c)** $x=\dfrac{3-p}{7-q}$
4.10	**(a)** $y+2=7x$ $x=\dfrac{y+2}{7}$ **(b)** $t^2=\dfrac{P}{5}$ $t=\sqrt{\dfrac{P}{5}}$ **(c)** $px-2x=q-5$ $x(p-2)=q-5$ $x=\dfrac{q-5}{p-2}$	**(a)** $x=\dfrac{y+2}{7}$ **(b)** $t=\sqrt{\dfrac{P}{5}}$ **(c)** $x=\dfrac{q-5}{p-2}$
4.11	**(a)** $y-8=3x$ $x=\dfrac{y-8}{3}$ **(b)** $r^2=\dfrac{S}{4\pi}$ $r=\sqrt{\dfrac{S}{4\pi}}$ **(c)** $4x-bx=6+a$ $x(4-b)=6+a$ $x=\dfrac{6+a}{4-b}$	**(a)** $x=\dfrac{y-8}{3}$ **(b)** $r=\sqrt{\dfrac{S}{4\pi}}$ **(c)** $x=\dfrac{6+a}{4-b}$

4.12	**(a)** $y+4=9x$ $x=\dfrac{y+4}{9}$ **(b)** $x^2=\dfrac{A}{6}$ $x=\sqrt{\dfrac{A}{6}}$ **(c)** $nx-2x=m+7$ $x(n-2)=m+7$ $x=\dfrac{m+7}{n-2}$	**(a)** $x=\dfrac{y+4}{9}$ **(b)** $x=\sqrt{\dfrac{A}{6}}$ **(c)** $x=\dfrac{m+7}{n-2}$
changing the subject		
4.13	$u^2=v^2-2as$ $u=\sqrt{v^2-2as}$	$u=\sqrt{v^2-2as}$
4.14	$2w=P-2l$ $w=\dfrac{P-2l}{2}$	$w=\dfrac{P-2l}{2}$
4.15	$2A=(a+b)h$ $h=\dfrac{2A}{a+b}$	$h=\dfrac{2A}{a+b}$
4.16	$y-c=mx$ $m=\dfrac{y-c}{x}$	$m=\dfrac{y-c}{x}$
4.17	$3V=\pi r^2h$ $h=\dfrac{3V}{\pi r^2}$	$h=\dfrac{3V}{\pi r^2}$
4.18	$3b^2=T-2a$ $b^2=\dfrac{T-2a}{3}$ $b=\sqrt{\dfrac{T-2a}{3}}$	$b=\sqrt{\dfrac{T-2a}{3}}$